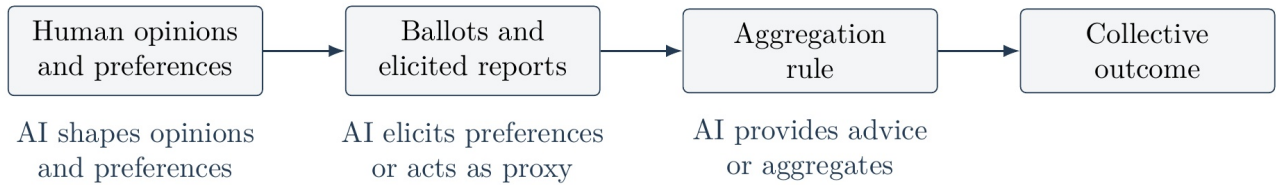


Foundations of Collective Decision Making Agents in a World of Artificial Intelligence

From elections and public budgeting to school placements and resource allocation, collective decisions shape everyday life. Each requires diverse individual interests to be translated into a shared outcome. Social choice provides a mathematical language for asking when this translation can be fair and efficient, and for proving when it cannot. Its axioms and quantitative guarantees raise fundamental questions about algorithms, approximation, incentives, communication and computational complexity. Classically, this process has three stages. Agents form opinions and preferences. Elicitation translates these into practical ballots, such as top choices or approvals. An aggregation rule then maps these reports to a collective outcome. Artificial intelligence can now intervene throughout this process: it can shape the information people receive, elicit preferences, act as a proxy, or advise on an outcome. Such systems can reduce the burden of expressing complex preferences, but they may misunderstand users, introduce bias, or make correlated mistakes. Classical assumptions therefore need to be re-examined under artificial mediation.



AI is changing markets. In auto-bidding, advertisers state high-level objectives while systems choose bids across auctions, changing equilibrium and welfare analysis compared to classical models [1]. The Bank of England likewise warns that shared AI models in trading may induce correlated behaviour and herding. These examples illustrate how individually useful artificial agents can alter collective outcomes.

Recent work studies LLM proxies [2], learning-augmented social choice [3], and social-choice approaches to alignment [4]. My programme asks a distinct question: what happens to classical guarantees when artificial agents intervene between human preferences and collective outcomes? I will trace how error, bias, strategic behaviour and computational limits propagate to welfare, fairness and representation, prove trade-offs when these goals are inconsistent, and design robust mechanisms for modern social choice.

Research Agenda

My research across algorithmic social choice, fair division, matching and game theory has equipped me with tools from approximation, complexity and axiomatic analysis. I am broadly interested in artificial mediation, but my main focus will be three connected directions.

Robust foundations. I will determine which classical guarantees survive when preferences or recommendations are incomplete, perturbed, biased or computationally restricted. The first objective is to characterise error propagation: if mediation at stage S introduces error ε_S , when does final loss grow smoothly with the ε_S in metric or utilitarian settings, and when is any such bound impossible? I will seek algorithms with tight degradation guarantees and matching lower bounds, beginning with voting and participatory budgeting. I will then study rules that receive both reported preferences and an AI recommendation, seeking an optimal trade-off between consistency when advice is accurate and robustness when it is unreliable. I will extend these guarantees to graph-structured objectives that capture complementarities among projects or committee members, and to repeated decisions with advice drawn from a shared distribution.

Preference formation, elicitation and delegation. Think of a less informed citizen choosing among unfamiliar candidates. They might describe their priorities in everyday language to an LLM and ask for alternatives to vote for. This is a direct influence, while AI can also shape their opinions indirectly by deciding which campaign ads appear in their social media feed. An AI agent can even mediate in an election, ask a few questions of each voter, and submit a ballot as a proxy. Classical social choice treats preferences as given before elicitation, but here the voter and AI jointly shape the submitted ballot. I will first assume fixed latent preferences and ask how accurately a proxy can recover the information needed for a good collective decision. In structured domains such as metric voting, I will characterise when $O(\text{polylog } m)$ adaptive queries can replace a full ranking over m alternatives while retaining a constant-factor welfare or representation guarantee. I will then consider AI-provided information that changes rather than merely reports preferences. Axioms for neutral advice, equal treatment and bounded influence will distinguish legitimate assistance from systematic manipulation. The resulting models will quantify how unequal model quality, strategic answers and shared bias affect query complexity and collective welfare. The final step is to design rules robust to biased AI influence on preferences.

Social choice for AI alignment. When an AI system affects several people, their conflicting judgments about its behaviour must be aggregated. Existing work develops axiomatic social-choice models for this problem [4]. I will study alignment when feedback is limited, noisy or unevenly distributed and the output space is too large to enumerate. How many adaptive comparisons are required to achieve welfare close to a full-information benchmark while satisfying unanimity or proportional representation? I will characterise communication–distortion and axioms–welfare trade-offs using approximation algorithms and lower bounds, and connect them to the composition results from the first direction.

Outcomes, feasibility and impact

I plan to pursue these directions in parallel through my network of collaborators around the world, focusing in years 1–2 on the development of the compositional framework and tight robustness bounds. Years 2–3 will address proxy elicitation and bounded influence, and years 3–5 will extend the theory to alignment and repeated decisions. Each project has a positive algorithmic or axiomatic target and a corresponding lower-bound or impossibility target, so informative results remain possible when the strongest guarantee fails. The core programme is theorem-driven and models artificial agents through general properties rather than assumptions tied to present-day LLMs. Where useful, simulations and real-world data will test modelling assumptions rather than replace mathematical guarantees. The resulting theory will provide principled ways to evaluate AI-mediated systems through the quality of their collective outcomes.

Selected references

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